WEEK 2 · LECTURE L2 · UNIT I

Maturity Models and Strategic Alignment

Where do we stand? What maturity models measure, why stage models are contested, the digital divide and the compliance frame, and why tools must follow strategy.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain what a maturity model is for, and the four claims every stage model makes.
2. Compare stage models (Nolan, the CMM, Kavanagh, Bersin) with the Westerman capability grid, and use the evidence against stage models.
3. Apply Henderson and Venkatraman’s Strategic Alignment Model: tools follow strategy.
4. Apply Kotter’s urgency step, the three level digital divide, and the DPDP and Social Security frames to an HR transformation.
5. Assess maturity across five dimensions and build a business case a CFO will accept.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 1 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

A consultant tells a firm it is at “level 2 of 5” in HR maturity and should move to level 3 next. What should the firm ask first?

A

How much level 3 costs

B

What the levels measure, what evidence supports the order, and whether level 3 fits the firm’s strategy

C

Which competitor is at level 4

D

Whether it can skip straight to level 5

EXAM LENS

The answer is B. This lecture gives you the theory behind each part of that question.

FRAME FIRST

What a maturity model is for, and what it measures

An argument, made legible

A maturity model makes an argument about capability understandable to people who do not share your expertise. Its purpose is prioritization and communication, not measurement in any strict sense.

What a score measures

How systematic, integrated and data driven a process is.

What it does not measure

Whether the process reaches every worker, whether leaders can steer it, or whether it is fair. That is why the course pairs a stage model with a capability grid and the digital divide lens.

THE CORRECT REPLY

“The model says we are at Level 2.” No: you said it, using the model.

THE FAMILY OF THEORY

What every stage model claims

Ordered

The stages form a sequence, not a menu.

Cumulative

Each higher stage contains more capability than the one below.

Non skippable

Each stage rests on the one below; buying a stage four capability without stage three foundations fails.

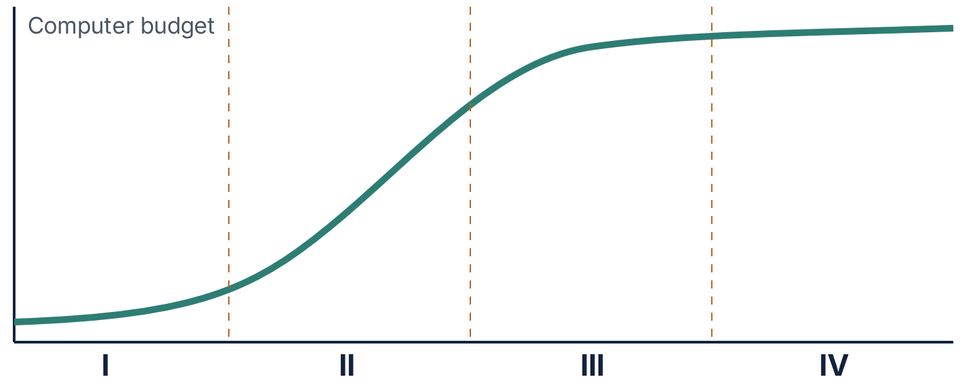
Diagnosable

An organization’s stage can be identified from observable indicators.

EXAM LENS

Non skippability is the strongest and most testable claim. Name it when asked to evaluate a stage model. Strength of the family: it gives a sequence, and explains failure when a stage is skipped. Weakness: one path and one pace for the whole organization.

What Nolan actually claimed



Stage	Budget pattern
I · Computer acquisition	Slow increases after acquisition
II · Intense system development	Highly increasing, often over 50% a year
III · Proliferation of controls	Increases slow, or the budget falls
IV · User/service orientation	Slow, even increases

EXAM LENS

From the computer budgets of three companies, Nolan read four stages off an S shaped spending curve. He offered the hypothesis “not as a prescription for computer management” but as a description, and hoped others would test it. The prescriptive “no skipping” use came later.

Redrawn schematically from Nolan (1973), Figure 2.

STAGE MODELS ARE CONTESTED

Later versions, and the evidence against them

The model was renamed

Gibson and Nolan (1974): initiation, expansion, formalization, maturity. Nolan (1979): six stages, initiation, contagion, control, integration, data administration, maturity. Cite the version you mean.

King and Kraemer (1984)

Weak evidence that organizations follow the stages; loosely defined variables; useful as a description, weak as a prediction.

Becker, Knackstedt and Pöppelbuß (2009)

Many maturity models lack documented development and are rarely tested before use; they propose a procedure for building them rigorously.

EXAM LENS

A maturity model is a structured opinion until its order and measures are tested. Use a stage to describe where an HR function is now, not to forecast where it must go next.

THEORY IN DEPTH · PAULK ET AL. (1993)

The Capability Maturity Model judges processes

The core idea

Built at the Software Engineering Institute, Carnegie Mellon, for software: maturity means repeatable, defined, measured processes.

How it works

Five levels: initial, repeatable, defined, managed, optimizing. Each rests on the one below.

Limits and critique

It can reward documentation over results, and suits stable processes better than fast changing ones.

IN PRACTICE

Recruitment run from a shared inbox is at the initial level: it works only because particular people know how.

EXAM LENS

The CMM is the origin of many HR maturity models; show you know where the ladder idea came from.

HR LADDERS · AS USED IN THIS COURSE

The HR technology maturity ladder, and where organizations stall

Stage	What it looks like
1 · Paper and spreadsheet (Nascent)	Records on paper or local files; manual reporting; intuition led; nobody owns the data
2 · Point solutions (Emerging)	A tool per problem; data siloed and rekeyed between tools
3 · Integrated (Defined)	One system of record with live feeds; dashboards; descriptive and diagnostic analytics
4 · Optimized (Advanced)	Processes redesigned around the data; predictive analytics and AI, governed for fairness
5 · Intelligent (Innovative)	Continuous experimentation; workforce planning shapes business strategy

EXAM LENS

The integration gap (2 to 3) is political: one definition of an employee must win. The capability gap (3 to 4) needs analysts and AI ethics skills. The culture barrier sits at every level. Functions sit at different levels at once.

Course framing after Kavanagh and Johnson (2018), with the five level framework of the course's Unit 1 teaching note (drawing on Kane et al., 2015).

BERSIN (2012)

The analytics ladder: four levels

Level	Question answered	What must be true first
1 · Operational reporting (reactive)	What happened? Counts and compliance reports	Headcount that agrees with itself
2 · Advanced reporting (proactive)	What is happening, and where? Dashboards, benchmarks	Scheduled, segmented, trusted reports
3 · Strategic analytics	Why is it happening? Segmentation, causes	Joins across systems and causal reasoning skills
4 · Predictive analytics	What will happen, and what should we do?	Deep, clean history and governance of predictions

EXAM LENS

Most HR functions sit at the first two levels, and skipping levels tends to fail. Kavanagh measures the plumbing; Bersin measures the thinking. An organization can be high on one and low on the other.

WESTERMAN, BONNET AND MCAFEE (2014)

The evidence behind the capability grid

Evidence base

Interviews with 150 executives in 50 large companies, then a survey of 391 companies with revenues of $500 million or more in 30 countries, with financial analysis of the 184 publicly traded. Technology firms and startups were deliberately excluded.

Two dimensions

Digital capability, the what: customer experience, operational processes, business models. Leadership capability, the how: a digital vision, engaging the organization, governance, technology leadership.

No bottom up masters

The authors found no successful bottom up transformations: every Digital Master was led top down, while engaging employees to build out the vision.

WESTERMAN ET AL. (2014)

Four types, and how they perform

Digital capability ↑

Fashionistas

+6% revenue · 24% less profitable. Buy every digital bauble, in incompatible silos

Digital Masters

26% more profitable · +9% revenue

Beginners

11% less profitable · 4% less revenue. Often cite regulation or privacy to wait

Conservatives

9% more profitable · 10% less revenue. Prudence becomes a governance trap

Leadership capability →

EXAM LENS

Digital capability goes with revenue; leadership capability goes with profit. The authors stress this is correlation, not proof of cause.

COURSE FRAMING

The leadership axis is the scarcer one. Firms buy their way along the digital axis because tools are procurable; governance, sequencing and political capital cannot be bought.

Figures against industry averages, from Westerman et al. (2014). Axes as in the digital mastery matrix.

Inside the two axes, and three traps for HR

Digital capability in HR

Employee experience (self service, mobile, chatbots); operational processes rethought, not merely automated; business model change, from reactive hiring to predictive planning.

Three HR traps (teaching note)

Technology culture gap: systems bought, spreadsheets still used. Pilot trap: champions make pilots work; scale needs orchestration. Disruption paradox: deliver now while transforming, so build governance and vision first.

Examples from the book

Asian Paints began as a Conservative: unified IT, centralized order taking, turned salespeople into connected relationship managers, automated plants. One manufacturer’s HR self service cut HR staff from over a hundred to under thirty while raising satisfaction.

The course’s Unit 1 teaching note adds “orchestration” as its gloss on coordination; teach the book’s four leadership capabilities.

CHOOSING THE INSTRUMENT

Stage models and capability grids

Question	Better instrument
Where are we, and what comes next in this domain?	Stage model: ordered levels, one path; shows the next step
Where should the marginal rupee go: capability or coordination?	Capability grid: independent dimensions, many paths
Why has our spending not produced results?	Grid: it diagnoses the Fashionista
What is the sequence for the next eighteen months?	Stage model
How mature is the organization, in one number?	Wrong question: maturity is a profile, not a score

EXAM LENS

A stage model gives a sequence but assumes an order the evidence may not support; a grid shows trade offs and imbalance but gives no single next step.

THE CRITIQUE

What maturity models cannot tell you

They assume one path

Sector, size and regulation may make a different route rational.

They describe, not explain

A model can place you at Level 2; it cannot say why, or whether climbing improves outcomes.

Higher is assumed better

For some organizations Level 3 is the efficient stopping point.

Many are vendor authored

Their highest level often resembles the vendor’s product.

Pöppelbuß and Röglinger (2011)

Maturity models serve three purposes: describing, prescribing, or comparing. A good descriptive model needs clear criteria; a prescriptive one also needs actions that improve the score. Many describe without prescribing, and few are tested.

EXAM LENS

Common mistake: averaging process scores into one maturity number. If payroll is at level 4 and people analytics at level 1, an average such as 2.2 describes no actual process: the spread is the finding.

RECENT EVIDENCE · THORDSEN, MURAWSKI AND BICK (2020)

Of 17 digital maturity models published from 2011 to 2019, most failed academic tests of valid measurement.

ASSESSING DIGITAL HR MATURITY

Five dimensions, and the weakest caps the whole

Dimension	What to look at
Technology infrastructure	Presence, integration, accessibility, reliability, scalability
Data capabilities	Quality, governance, analytical capability, data driven culture
User experience	Ease of use, personalization, self service, satisfaction
Organizational capability	Digital literacy, analytical skills, change capability, innovation culture
Governance and strategy	Alignment, decision rights, investment, risk management

EXAM LENS

Methods: triangulate quantitative measures, qualitative evidence of how managers really decide, benchmarking, and self assessment, which tends to overstate maturity. Read patterns, not averages; find the limiting factor; set targets by context; advance in stages.

Course’s Unit 1 teaching note, drawing on DeLone and McLean (2003), Wang and Strong (1996) and Weill and Ross (2004).

THEORY IN DEPTH · HENDERSON AND VENKATRAMAN (1993)

The Strategic Alignment Model

Business strategy

Business scope · distinctive competencies · business governance

← functional integration →

IT strategy

IT scope · systemic competencies · IT governance

↕ strategic fit

↕ strategic fit

Organizational infrastructure and processes

Administrative infrastructure · processes · **skills**

← functional integration →

IT infrastructure and processes

Architecture · processes · **skills**

Four domains with three dimensions each (twelve in all). Strategic fit links external choices to internal arrangements; functional integration links business to technology. Alignment needs both.

HENDERSON AND VENKATRAMAN (1993) FOR HR

Tools follow strategy

Core claim

Technology creates value only when it fits the business strategy and the organization’s capability to use it. Misalignment, not poor technology, is the usual reason investments fail.

Read for HR

Business strategy: the firm’s strategy. IT strategy: the HR technology strategy. Organizational infrastructure: HR roles and processes. IT infrastructure: the actual systems.

Skills: the HR hook

Skills appear in both infrastructure domains: the people who carry out the strategy, and the people who run the systems. A misalignment in either is an HR problem.

EXAM LENS

Three kinds of relationship: a bivariate fit links two domains; a cross domain alignment runs through three, in one direction; strategic alignment covers all four. Each cross domain view leaves one domain unexamined. A retailer growing through frontline staff in smaller towns, whose HR systems serve only head office, is misaligned.

FOUR ALIGNMENT PERSPECTIVES

What drives what

Perspective (1993 article)	Name in the 1990 working paper	Chain
Strategy execution	Strategy implementation	Business strategy → organizational infrastructure → IT infrastructure
Technology transformation	Technology leverage	Business strategy → IT strategy → IT infrastructure
Competitive potential	Technology exploitation	IT strategy → business strategy → organizational infrastructure
Service level	Technology implementation	IT strategy → IT infrastructure → organizational infrastructure

EXAM LENS

Strategy implementation is the most common, matching the top down view of strategy; excitement about technology exploitation without redesigning internal operations is seriously limited. A six step process: clarify strategy; translate into workforce implications; identify capability gaps; prioritize interventions; build business cases; execute and measure.

Names matched by direction from the course's Unit 1 teaching note and the 1990 MIT working paper (CISR WP No. 217).

STRATEGIC ALIGNMENT

Alignment is a journey, and a module list is not a strategy

The outcome test

What business number moves, by how much, by when?

The ownership test

Who outside HR is accountable for that number?

The sacrifice test

What are we stopping in order to fund this?

The reach test

Which workforce segment does this actually touch?

EXAM LENS

Common mistake: treating alignment as a one time check at purchase. Henderson and Venkatraman call it a dynamic concept, like shooting at a moving target; one manager called it “a journey and not an event”. A sequenced list of purchases with no stated business outcome is not a strategy; apply all four tests (Course framing).

RECENT EVIDENCE · WESSEL, BAIYERE, OLOGEANU-TADDEI, CHA AND BLEGIND JENSEN (2021)

Two long case studies separate the two: digital transformation uses technology to redefine what the organization offers; IT enabled transformation uses it to support what the organization already offers. Ask which one a proposed HR system does.

KOTTER (1995; 1996)

Eight steps, derived from failures

Create the climate

1. Establish urgency. 2. Form a guiding coalition. 3. Develop a vision and strategy.

Engage the organization

4. Communicate the vision. 5. Empower broad based action. 6. Generate short term wins.

Sustain the change

7. Consolidate gains. 8. Anchor the change in the culture.

EXAM LENS

Kotter (1995) estimated that about three quarters of management must honestly believe the status quo is more dangerous than change before urgency is real. Distinguish true urgency (evidence based) from false urgency (anxious activity). He derived the steps from where failures clustered, which is also their limit: descriptive of failure, not tested as a recipe.

VAN DIJK (2005); VAN DEURSEN AND HELSPER (2015)

The three level digital divide

Level	Question	Remedy
1 · Access	Can the person physically reach the system: device, connectivity, data cost, interface language?	Design and procurement; training cannot fix a feature phone
2 · Skills	Given access, can they operate it confidently, without fear of visible mistakes?	Capability building, in their language, in paid time
3 · Outcomes	Given access and skill, do they obtain the benefit: the entitlement, the payment, the job?	Redesign the process so the benefit actually arrives

EXAM LENS

Each level needs a different remedy, and an average conceals all three. In India the divide often runs inside the firm, between corporate offices and the frontline.

RECENT EVIDENCE · LYTHREATIS, SINGH AND EL-KASSAR (2022)

A review of 50 studies from 2017 to 2021 finds education the factor most linked to the divide and most research now at the second level, skills. It names two possible new levels: awareness of algorithms, and inequality in data.

THE COMPLIANCE FRAME

DPDP Act 2023 and the Code on Social Security 2020

DPDP Act 2023

The employee is a Data Principal; the employer is the Data Fiduciary and answers for every vendor. Notice, purpose limitation and correction rights become system requirements, not just policies.

Code on Social Security 2020

Gives gig and platform workers legal definitions and places welfare contributions on aggregators (1 to 2 percent of annual turnover, capped at 5 percent of amounts paid to these workers) without making them employees.

EXAM LENS

Legal obligations precede the business case: they set requirements no cost benefit argument can override.

BUILDING THE BUSINESS CASE

Eight components and three numbers a CFO will check

Eight components

1. Strategic context and problem
2. Proposed solution
3. Benefits: efficiency, effectiveness, strategic
4. Full costs, initial and ongoing
5. ROI
6. Risks and mitigation
7. Implementation roadmap
8. Success metrics, leading and lagging

Number	Formula
Simple ROI	(total benefits − total costs) ÷ total costs
Payback	Time until cumulative benefits equal cumulative costs
NPV	PV of benefits − PV of costs, at the cost of capital

FROM THE TEACHING NOTE

Four kinds of value: financial, operational, strategic and risk. Use conservative estimates; best, base and worst cases; staged investment.

APPLY

A maturity score for sale

Teaching case (Course framing)

A vendor gives a Chennai hospital group a “digital HR maturity score” of 2.3 out of 5 and proposes a three year roadmap to reach 4, beginning with an AI analytics module. The group’s strategy is to open 20 small town hospitals and staff them with locally hired nurses.

Work it through

1. What would King and Kraemer ask about the score?
2. Is the roadmap a stage model or a grid, and what does that assume?
3. Where does the analytics module sit on Bersin’s levels, and is the group ready?
4. What would Henderson and Venkatraman say should come first?

APPLY

Sequencing out of the Fashionista corner

1 · Pause

New tool purchases

2 · Vision and decision rights

Who owns the record; who owns analytics

3 · Governance

For data and AI

4 · Rationalize and integrate

The existing stack, before adding to it

COURSE FRAMING

The self critique: this delays visible wins, so it needs one quick, credible improvement for frontline staff to keep the Board’s patience. This answers the Lesson Plan question: high on technology, low on leadership capability, what sequence?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A higher maturity level is always the goal	Maturity matters only in relation to strategy
Stage models describe how organizations really develop	Evidence for fixed stages is weak (King and Kraemer, 1984)
Strategic alignment means buying the best tools	It means matching strategy, processes, skills and technology
One maturity number sums up the function	Maturity is a profile; the spread is the finding
Alignment is checked once, at purchase	Alignment is continuous: a moving target

COMPARE AND EVALUATE

Eight lenses on where an organization stands

Theory	Use it to	Main weakness
Nolan (1973)	Describe a growth path	Weak empirical support (King and Kraemer)
CMM (Paulk et al., 1993)	Judge process discipline	Can reward documentation over results
Bersin (2012)	Place analytics capability	A consultancy model, little tested *
Westerman et al. (2014)	Diagnose imbalance	Correlation, not cause; no sequence
Pöppelbuß and Röglinger (2011)	Ask what a model is for	Design principles, not a model itself *
Henderson and Venkatraman (1993)	Test fit between strategy and technology	Alignment can become rigidity
Kotter (1995)	Sequence the change	Derived from failures, not tested as a recipe
Three level divide	Locate exclusion	Remedies need local evidence *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

An Indian retailer has been rated “level 2” on an HR maturity model. Using the critique of stage models, Westerman’s grid and the strategic alignment model, advise the CHRO on how to use, and not misuse, the rating. **20 marks**

A strong answer: explains stage models and their weak empirical support; contrasts a stage model with a grid; applies strategic fit and functional integration to the next step; recommends how to use the rating, and notes a limit.

“Large Indian organizations do not have a maturity level; they have a maturity gap.” Using stage models and the Westerman grid, critically evaluate the usefulness of maturity models for HR. **20 marks**

A strong answer: explains the four claims of stage models; contrasts Kavanagh and Bersin; applies the grid; uses King and Kraemer and Pöppelbuß and Röglinger; argues maturity is a profile, not a score.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Explain the three level digital divide and show why each level requires a different remedy in an Indian organization. **20 marks**

A strong answer: defines access, skills and outcomes; gives a remedy per level; explains why averages conceal the gap; links to the DPDP Act and frontline realities.

What are Kotter’s eight steps, and why does he argue the first step decides the rest? **10 marks**

A strong answer: lists the steps in three phases; explains true versus false urgency and the three quarters threshold.

A business case reports a simple ROI of 150 percent. What two further numbers should a CFO ask for, and why? **10 marks**

A strong answer: payback period and NPV, with their formulas; notes that ROI ignores timing and the cost of capital.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **A maturity model is a sequencing discipline, not a destination.**
   Use it to argue priorities, not to report a score; state which purpose it serves.

2. **One number hides more than it shows.**
   Kavanagh measures the plumbing, Bersin the thinking, Westerman the capacity to direct either.

3. **Tools follow strategy.**
   Alignment needs strategic fit and functional integration, maintained over time.

4. **Exclusion and obligation are diagnoses, not afterthoughts.**
   Access, skills and outcomes need different remedies; the law sets requirements no business case can override.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Westerman, G., Bonnet, D., & McAfee, A. (2014). *Leading digital*, the maturity framework.
- Deloitte, Global Human Capital Trends (latest), selected sections.

NEXT WEEK

- Johnson, Carlson and Kavanagh (2025), Chapter 11; Chen, Chiang and Storey (2012); Angrave et al. (2016).
- Manuti and de Palma (2018), Chapters 3 and 6; the DPDP Act 2023 orientation note.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

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